

Examiner reports

Q1.

In this question part (a) was very well answered, most candidates succeeding in obtaining the printed answer correctly. The conditional probability in part (b) was found correctly by a good number of candidates, but others substituted numbers very inappropriately into a formula and obtained no credit in this part of the question.

Q2.

Most candidates showed a good grasp of basic probability in part (a) of this question. Some incorrectly used subtraction from 1 in part (a)(ii) and some misunderstood the phrase 'exactly one', confusing it with 'at least one', in part (a)(iii). A small proportion of candidates appeared to be confused by the use of percentages in the question.

A good number of candidates were able to earn at least one mark in part (b) by using their answer to part (a)(iii) as the denominator of a fraction, though the numerator was less often correct.

Q3.

Part (a) of this question offered a good number of marks to most candidates. A direct approach using the numbers in the table was more successful than a slavish dependence on formulae, to some extent in part (a)(iii) but even more so in part (a)(v). In this latter part the formula relating to the union of two sets was often used without excluding the intersection as required.

The same considerations applied in part (b)(i), where the use of a formula for conditional probability often led to incorrect calculations, whereas counting the number of British players and British goalkeepers usually gave rise to the correct answer.

In part (b)(ii) most candidates earned no marks. The comments were often related to general footballing considerations rather than to the 30 players referred to in the question, or candidates confused the word 'independent' with the phrase 'mutually exclusive'. It was hoped that the candidates would see that the original probability of the chosen player being a goalkeeper was altered given the information that the chosen player was British, so that the given events could not be independent.

Q4.

The majority of candidates drew a tree diagram here and were then able to write down their results confidently. After the two multiplications in parts (a)(i) and (a)(ii), it was quite common to see a third multiplication in part (a)(iii) instead of an addition of the first two answers. Conditional probability continues to be a source of difficulty for many candidates, but there were several instances of good work in part (b) from candidates who had trouble with other questions on the paper.

Q5.

Many candidates tackled part (a) of this question on probability by drawing a clear tree diagram and then giving correct values for all the required probabilities.

Part (b) was found considerably more difficult, though an encouraging number of candidates showed enough knowledge of conditional probability to earn partial credit.

Q6.

The performance of the candidates on this question was, as often with questions on probability, of variable quality. Most candidates seemed to find parts (a)(i) and (a)(ii) straightforward, but the request in part (a)(iii) was not well answered - a common mistake being to consider the probability of 'exactly one' successful outcome instead of 'at least one' successful outcome.

This, in turn, caused some loss of marks in part (b), where most candidates showed no true understanding of conditional probability. Some candidates found themselves trying to divide 0.9 by 0.66. The examiners could see how this had arisen, but no credit could be given for a value greater than 1 for a probability.

Q7.

Many candidates scored full marks in part (a). The vast majority were able to draw a correct tree diagram and most realised they had to multiply out the two branches and add these probabilities.

Errors occurred mainly in the multiplication of decimals. Few candidates scored well in part (b) with a significant proportion of candidates not attempting this part. A common error made by those attempting part (b) was to use 0.8 as the denominator, that is, using the probability of being inoculated rather than the probability of being immune. The topic of conditional probability did not seem to be well understood by candidates.

Q8.

Part (a) was usually answered correctly, though a minority of candidates worked out the probability in the first branch of the tree, neglecting the third branch. Very few candidates showed any knowledge of conditional probability in part (b).

Q9.

Nearly all the candidates succeeded in finding the probabilities asked for in part (a) of this question and a remarkably high proportion coped well with the conditional probability in part (b), simply dividing the answer to part (a) (i) by their answer to part (a) (ii).

Q10.

Part (a) was usually well answered.

Many candidates used vague theoretical arguments to answer part (b) rather than comparing the probabilities calculated in parts (a)(i) and (a)(iii). The rest of the question was generally well answered.

Q11.

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Q13.

This was reasonably well answered. In part (b), a surprisingly large number of candidates selected with replacement and/or omitted the 2 in part (b)(ii).

Most were successful with part (c)(i) but in part (c)(ii) few evaluated the necessary probabilities, many preferring to argue, incorrectly, that R and S were independent because it was possible for a member to be both female and aged under 30.

Q14.

Most candidates scored full marks in parts (a)(i) and (ii), but many found parts (iii) and (iv) hard and did not seem able to cope with conditional probabilities. A large number of candidates answered part (b) confidently, gaining 1 mark, but few expressed the probability simply, and without negatives, which was required for 2 marks.

Q15.

Some perfect solutions were seen but these were few and far between. Most candidates struggled with this probability question and often seemed to multiply or add any available probabilities, frequently ending up with a solution that was greater than 1. Many candidates simply did not attempt this question.

Q16.

Most candidates were able to make a good attempt at all but part (c)(iii). However, $\frac{3}{150}$ was not an uncommon answer to part (a)(ii), and in part (b) the 6 was sometimes omitted or replaced by 3. Part (c)(iii) caused problems. Attempts to evaluate the probabilities of all possible outcomes of the first two letters became unmanageable, compounded in some cases by unnecessarily considering the third letter as well. The only successful method was to consider the probability of each of the first two letters not being "suitable for publication from a writer who is for the proposal".

Q17.

Several perfect solutions were seen from candidates who understood the laws of probability but most candidates struggled with this probability question and often missed the point that the events involved were not independent. A significant number of candidates did not attempt the question.

Q18.

Many candidates scored full marks in parts (a)(i) and (ii) but several found parts (iii) and (iv) hard and assumed independence of events when this was not true. Many found the conditional probabilities in parts (v) and (vi) difficult but there were a considerable number

of fully correct solutions. A large number of candidates answered part (b) correctly but some were inaccurate in their description of the event in simple terms. Some stated that the event $(R \cap H \cap T)$ referred to a male who is aged 55-74 or who has high blood pressure or who is being treated, despite obtaining the correct answer for $P(R \cap H \cap T)$.

Q19.

Although most candidates were able to answer parts (a) and (b), few were able to deal with the given conditions in the other parts. Many, for example, in part (c) calculated the probability of Louise winning the first game and winning the series by three games to one, instead of taking Louise's winning the first game as given.

Q20.

This challenging probability question was exceptionally well answered by a significant number of candidates. There were many candidates scoring full marks.

The proof in part (a) was sometimes done by candidates' assuming both 0.28 and 0.2 to be true rather than by showing that if $P(B) = 0.1$ then $P(C) = 0.2$. Also some candidates showed that if $P(C) = 0.2$ then $P(B) = 0.1$, which was allowed.

Part (b) was not always correctly answered but most candidates showed a good understanding of conditional probability.

Q21.

Candidates had few problems with this question.

Q22.

Many candidates scored highly on this question with a substantial number scoring full marks. Some candidates still avoided the probability question, but these were few and far between. The only difficulty came with part (d), where clearly some candidates had not met the idea of conditional probability.

Q23.

Many candidates scored highly on part (a) of this question, with a significant number scoring full marks. However, many then found the extension in part (b) too complicated to attempt.

Q24.

A common mistake on part (a)(iii) was to multiply the probability of choosing a junior member by the probability of choosing a female member. Part (b) proved testing. A good proportion were able to determine whether or not events were mutually exclusive. Those who attempted to explain why V and W were not independent usually gave an explanation in rather vague, general terms, rather than the required numerical demonstration.

Q25.

Many candidates scored full marks in parts (a), (b) and (c) but most found the conditional probability in part (d) difficult to find. A surprising number failed to get the correct answer to parts (a) and (b) but answered part (c) correctly!

Q26.

Most errors in this question were small slips rather than errors of method.

Q27.

Many candidates scored full marks in part (a) but found the event in part (b) difficult to express in simple terms.

Q28.

Many candidates scored full marks and most made a very good effort. There were still a few candidates who clearly avoided questions on probability.

Q29.

Most candidates scored highly on this question; they were evidently comfortable tackling probability questions.

Q30.

It was encouraging that many students did well on this question. However, some candidates couldn't manage part (a)(ii) or (iii) yet gave a correct answer to part (a)(iv).

In part (b), very few correct solutions were given, as answers were not defined in the context of the question. Many simply stated that it was the union of B and D, and those who did refer to the two events rarely mentioned that it was 'either' or 'both' events.

Q31.

Most candidates scored highly on this question and part (b) produced many good solutions, with few candidates having problems.

Q32.

There was a good spread of marks in this question, with the usual mistakes of using 'with replacement' in part (b) and omitting the 3 in part (b)(iii).